Laws of Indices

Exam code: 0607

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Laws of indices

What are the laws of indices?

- **Index laws** are **rules** you can use when doing **operations** with **powers**
  - They work with both **numbers** and **algebra**

Law	Description	How it works
$a^1 = a$	Anything to the power of 1 is itself	$6^1 = 6$
$a^0 = 1$	Anything to the power of 0 is 1	$8^0 = 1$
$a^m \times a^n = a^{m+n}$	To multiply indices with the same base, add their powers	$4^3 \times 4^2$ $= (4 \times 4 \times 4) \times (4 \times 4)$ $= 4^5$
$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$	To divide indices with the same base, subtract their powers	$7^5 \div 7^2$ $= \frac{7 \times 7 \times 7 \times \cancel{7} \times \cancel{7}}{\cancel{7} \times \cancel{7}}$ $= 7^3$
$(a^m)^n = a^{mn}$	To raise indices to a new power, multiply their powers	$(14^3)^2$ $= (14 \times 14 \times 14) \times (14 \times$ $= 14^6$
$(ab)^n = a^n b^n$	To raise a product to a power, apply the power to both	$(3 \times 4)^2 = 3^2 \times 4^2$

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$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	To raise a fraction to a power, apply the power to both the numerator and denominator	$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$
$a^{-1} = \frac{1}{a}$ $a^{-n} = \frac{1}{a^n}$	A negative power is the reciprocal	$6^{-1} = \frac{1}{6}$ $11^{-3} = \frac{1}{11^3}$

How do I deal with different bases?

- Index laws only work with terms that have the **same base**
  - $2^3 \times 5^2$ cannot be simplified using index laws
- Sometimes expressions involve different base values, but **one is related to the other by a power**
  - e.g. $2^5 \times 4^3$
- You can use powers to **rewrite one of the bases**
  - $2^5 \times \mathbf{4}^3 = 2^5 \times (\mathbf{2}^2)^3$
  - This can then be simplified more easily, as the two bases are now the same
  - $2^5 \times (2^2)^3 = 2^5 \times 2^6 = 2^{11}$

Worked Example

(a) Find the value of x when $6^{10} \times 6^x = 6^2$

Answer:

Using the law of indices $a^m \times a^n = a^{m+n}$ we can rewrite the left hand side

$$6^{10} \times 6^x = 6^{10+x}$$

So the equation is now

$$6^{10+x} = 6^2$$

Comparing both sides, the bases are the same, so we can say that

$$10 + x = 2$$

Subtract 10 from both sides

$$\mathbf{x = -8}$$

(b) Find the value of n when $5^n \div 5^4 = 5^6$

Answer:

$$5^n \div 5^4 = 5^{n-4}$$

So the equation is now

$$5^{n-4} = 5^6$$

Comparing both sides, the bases are the same, so we can say that

$$n - 4 = 6$$

Add 4 to both sides

$$n = 10$$

(c) Without using a calculator, find the value of 2^{-4}

Answer:

Using the law of indices $a^{-n} = \frac{1}{a^n}$ we can rewrite the expression

$$2^{-4} = \frac{1}{2^4}$$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$ so we can rewrite the expression

$$\frac{1}{2^4} = \frac{1}{16}$$

$$\frac{1}{16}$$

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